

Education

Year	Degree	Institution	Performance
2025 - Present	M.Sc. Statistics	Indian Institute of Technology Kanpur	8.66
2025	B.Sc. Statistics (Hons.)	The Maharaja Sayajirao University of Baroda	8.74
2021	Class-XII (CBSE)	P. P. Savani School, Ankleshwar	94.4%
2019	Class-X (CBSE)	P. P. Savani School, Ankleshwar	97.4%

Internship

- Research Intern | Student Advance Research Internship Program, IIT Kanpur

(May'26 - Jul'26)
- Mentor: Dr. Abhishek Kumar Gupta, Department of Electrical Engineering, IIT, Kanpur
- Project: Detection of Clustering in Stochastic Transmitter Processes via Molecular Communication Signals

Objective	• To devise a statistical test for detecting clustering in transmitter arrangements in a 3D molecular comm. system, using only the aggregate received signal $Y(t)$, with no knowledge of individual transmitter locations.
Approach	• Modelled transmitter spatial distributions in 3D using homogeneous Poisson Point Process (PPP) and inhomogeneous PPP (iPPP) with exponentially decaying intensity (β), around a absorbing spherical receiver. • Derived molecule hitting probabilities using Fick's diffusion equation and modeled the received signal $Y(t)$ as a sum of independent Binomial random variables across transmitters. • Simulated $Y(t)$ across parameter β , identifying peak height, peak time, and rise time as clustering indicators. • Developed framework for the distribution of $Y(t)$ for an ongoing Fisher-Information-based clustering detection test.
Impact	• Established a foundation for signal-only clustering detection , with relevance to drug delivery and neuroscience.

Key Projects

- Mortgage Credit Risk Analytics |  | Self Project

(Oct'25 - Mar'26)
- Developed a mortgage credit risk framework on Freddie Mac data (33M+ performance records) to model default risk.

– Built a credit risk pipeline using sentinel-value cleaning, WOE/IV encoding, imbalance handling, vintage splits on features.

– Engineered behavioural and macroeconomic features to benchmark **XGBoost, Random Forest, and Logistic Regression**.

– Selected XGBoost for optimal performance, achieving a **0.12 Brier score** and **0.74 PR-AUC** on test data.

– Analysed vintage risk using **PSI** and delinquency curves, distinguishing population shifts across benign and crisis vintages.

– Estimated LGD using a **2-stage GLM** on 17K+ defaulted loans, and calculated Expected Loss from **PD, LGD, and EAD**.

– **Stress-tested** PD, LGD, and EL across crisis vintages, PD rose 2.06× and EL increased from 3.94% to 8.50% of exposure.
- Vehicle Damage Severity Analysis |  | Self Project

(Jun'26 - Jul'26)
- Developed a **multi-label** deep learning model in **PyTorch** to classify **6 vehicle damage types** using the CarDD dataset.

– Built a computer vision pipeline with **COCO annotation parsing, image augmentation**, and ImageNet normalization.

– Compared **custom CNN, VGG16, and ResNet50** models, addressing **class imbalance** across 2,816 training images.

– Achieved a best **72.2% Macro F1-score** with VGG16, outperforming custom CNN (58.2%) and ResNet50 (69.6%).
- Yield Curve Forecasting |  | Self Project

(Jan'26 - Feb'26)
- Built a Treasury yield-curve forecasting framework using **30+ years of US yields and macro indicators**.

– Tested stationarity using **ADF & KPSS** tests, confirming yields as **I(1)**, and used **ACF & PACF** to identify ARIMA orders.

– Benchmarked Random Walk, **ARIMA, ARIMAX, VAR, and Dynamic Nelson-Siegel** models via walk-forward forecasts.

– Achieved up to **32% lower RMSE** than Random Walk and built an **interactive dashboard** for yield forecasting and pricing.
- FilingIQ |  | Self Project

(Apr'26 - Jun'26)
- Built a **RAG pipeline** over annual filings of **JPMorgan, Bank of America, and Goldman Sachs** for financial analysis.

– Processed **10,870 text segments**, generated **all-MiniLM-L6-v2** embeddings, and indexed them in **FAISS** for retrieval.

– Designed a **RetrievalQA chain** using a **Groq-hosted LLM** via **LangChain**, retrieving top-4 grounded chunks per query.

– Developed a **Streamlit UI** with source attribution and validated answers against financial metrics from source filings.
- longevityRiskR |  | Course Project

(Feb'26 - Mar'26)
- Mentor: Prof. Dootika Vats, Dept. of Statistics and Data Science | MTH209 - Data Science Lab II

– Developed an **R package** implementing a **stochastic mortality model** with exponential improvement and Gaussian shocks.

– Built **Monte Carlo annuity valuation** functions comparing **expected-value and quantile-based** premium pricing.

– Designed a **ruin-probability engine** to quantify **insolvency risk** under stochastic mortality and insurer payout assumptions.

– Implemented **survival probability and life expectancy** estimators from simulated mortality paths with risk visualizations.
- Energy Consumption Prediction |  | Self Project

(Nov'25 - Dec'25)
- Built an end-to-end energy consumption pipeline with **preprocessing, categorical encoding, and feature selection**.

– Assessed OLS assumptions using **VIF, Breusch-Pagan, Durbin-Watson, Jarque-Bera** and residual diagnostics.

– Compared **OLS, Ridge, Lasso, and Elastic Net Regression** using **TimeSeriesSplit** and forward validation.

– Selected **Elastic Net** as the best-performing model, achieving **80.8% test R^2** on a chronological 70/30 holdout.
- General Insurance: Awareness & Preferences |  | Course Project

(Jul '24 - Apr '25)
- Mentor: Dr. Mangala Shah, Department of Statistics, The Maharaja Sayajirao University of Baroda

– Conducted a **primary survey** using **4-stage area sampling** to assess General Insurance awareness in Vadodara.

– Designed a **structured questionnaire** quantifying insurance literacy, purchase behavior, and product preferences.

- Applied **non-parametric & association tests (Wilcoxon, Chi-Square, etc.)** in **R** to analyze demographic impact.
- Identified income, education, and employment as key drivers of insurance uptake; affordability and unawareness as barriers.
- **Customer Segmentation** | 🧠 | *Self Project* (Feb’26 - Mar’26)
 - Developed a **customer segmentation strategy** from 2,000+ records, handling missing values and outliers.
 - Applied **PCA** to reduce 29 features to 2-3 principal components for visualization and analysis.
 - Implemented **K-Means, Hierarchical, and Gaussian Mixture** clustering, achieving a **0.71 silhouette score**.
 - Identified **3 customer segments**, projected to increase engagement by **20%**.
- **Monetary Policy and Market Volatility** | 🧠 | *Self Project* (May’26 - Jun’26)
 - Built an **NLP** market-risk pipeline using RBI policy documents (2020-25) with NIFTY 50 and India VIX to predict volatility.
 - Engineered **TF-IDF** and 100D **GloVe** text signals with lagged returns and momentum features under chronological alignment.
 - Developed **Random Forest, RNN, and LSTM** models predicting India VIX shocks across market and NLP features.
 - Constructed document-level policy representations to quantify themes across inflation, growth, liquidity, and financial stability.
- **ControlPlane.ai: Real-Time AI Governance & Risk-Control Layer** | 🧠 (Aug’26 - Ongoing)
Accenture Innovation Challenge 2026
 - Building a real-time control layer that scores every **LLM** response for performance, cost, and responsibility risk before delivery.
 - Designing **Brier, self-consistency, PSI, and KS** monitoring to detect hallucination, miscalibration, and distribution drift.
 - Implementing a tiered **allow/edit/review/block** policy with tuned thresholds under **asymmetric FP/FN costs**.
 - Developing a **Python** prototype with audit trails logging **flag rates, override frequency**, and decision outcomes per tier.

Scholastic Achievements

- Qualified **Joint Admission Test for Masters (JAM) Mathematical Statistics 2025** among 2880 candidates.
- Qualified **Graduate Aptitude Test in Engineering (GATE) Statistics 2025** among 1967 candidates.
- Awarded the **Prof. Ashok Manilal Praikh Gold Medal** for securing the highest CPI in B.Sc. Statistics (Hons.), The Maharaja Sayajirao University of Baroda, 2025.
- Qualified for the final round of the **AI Hackathon** under **SARCathon 2025**, organized by the Student Alumni Relations Cell, IIT Bombay.
- Granted the **National Board Higher Mathematics (NBHM) Scholarship 2025** for pursuing Masters’ Degree in Statistics.
- Cleared 4 actuarial papers- **CM1, CS1, CB1, and CB2** from the **Institute of Actuaries of India**.
- Secured **AIR-2** in the **13th Statistics Olympiad, 2023**, by CR RAO AIMSCS, among participants from India and Sri Lanka.

Position of Responsibility

- Secretary, Academics and Career Council, IIT Kanpur** (2025-26)
- Appointed as the Secretary of the **Academics Wing** in the AnC council, representing the PG students on academic matters.
 - Member of the Technical Team for the **Institute Research Symposium 2025**, contributing to the Schedule and Abstract Book committee.
 - Contributed to the **“You Are Not Alone” initiative**, collecting anonymous student concerns about academic and research-related issues and constructively communicating them to the institute administration.

Technical skills

- **Programming Languages:** Python, R, SQL
- **Libraries & Packages:** Numpy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-Learn, Statsmodels, ggplot2, rvest, dplyr, stats
- **Utilities:** MS Office, L^AT_EX, R Markdown

Relant Courses

*Ongoing

Data Science Lab 1, 2	Non-Linear Regression*	Statistical Inference	Statistical Computing
Time Series Analysis	Game Theory*	Stochastic Processes	Financial Economics

Extra Curricular Activities

- Volunteered for the **FOSTER 2023**, the annual science Open House of the Department of Statistics, The Maharaja Sayajirao University of Baroda.